claude-wsl-ubuntu / 4f364eb7-a600-4a2c-9b37-e5a313b6db9d

Metadata

- Source: `claude-wsl-ubuntu`
- Kind: `claude`
- Source file: `\\wsl.localhost\Ubuntu\home\avidullu\.claude\projects\-home-avidullu-projects-khelsutra-guru\4f364eb7-a600-4a2c-9b37-e5a313b6db9d\subagents\workflows\wf_b0aa0abe-638\agent-a354d35dc4b2c2b42.jsonl`
- SHA-256: `de8ab4b5fee1283abfd1dce659a9e6f7c14b5eaefbf38078e5d6177c76e19cdf`
- Source modified: `2026-06-29T17:03:55+00:00`
- Imported at: `2026-07-05T16:48:30+00:00`
- project: `wf_b0aa0abe-638`
- session_id: `4f364eb7-a600-4a2c-9b37-e5a313b6db9d`

Transcript

1. user (2026-06-29T17:01:41.471Z)

```
In /home/avidullu/projects/khelsutra-guru/badminton-highlight-indexer, read docs/DOC_STATUS.md
IN FULL (§1 legend, §2 archival due-diligence checklist, §3 register, §4 open findings). Then
determine which tracked docs are now TERMINAL (DONE/REJECTED) and archive-ready vs still
LIVE/gated.
Specifically assess: docs/RALLY_DETECTION_FIXES_PLAN.md (now Status: DONE ? all 3 findings A/B/C
resolved, P2b merged #396): is it archive-ready per DOC_STATUS, and to which archive subdir
(docs/archives/past_projects/ ?). Also assess the R-series rally-detection quality cluster ?
RALLY_DETECTION_QUALITY_REPORT.md, RALLY_DETECTION_CODE_REVIEW.md, QUALITY_ITERATIONS.md,
R2_EVAL_METRIC_DESIGN.md, RALLY_QUALITY_RESEARCH.md, and any other RALLY_DETECTION_* / quality
docs ? which are terminal vs gated (DOC_STATUS notes the cluster is gated until R2 tolF1
promoted/rejected as the gate + §7.1 single quality number attributed; P2b promoted the flag ?
does that close the gate or not?).
For EACH archive-ready doc list: current Status, every INBOUND reference (grep the repo for the
exact filename across docs/*.md, docs/README.md, CLAUDE.md, NEXT_STEPS.md, cross-doc links) as
file:line, and the §2 steps to run before git mv. BE CONSERVATIVE ? only flag genuinely terminal
docs; for the rest give the precise gate. The owner explicitly wants all COMPLETED docs moved to
archives/, so be thorough about what truly qualifies.
```

2. assistant (2026-06-29T17:01:43.837Z)

```
I'll start by reading the DOC_STATUS.md file in full, then investigate the specific docs
mentioned.
```

3. user (2026-06-29T17:01:44.644Z)

```
1       # Documentation Status & Archival Register
2
3       > **Status:** ? **LIVING** ? the single board for doc **lifecycle + archival due-
diligence**. **Last full review:** 2026-06-21.
4       > **Update cadence:** whenever a doc changes lifecycle state (created / shipped a
milestone / delivered / archived), and re-review at each clean-slate.
5       > **Why this exists:** so we can (a) **review completeness** at a glance, (b) **update
it as work finishes** to prove we're done, and (c) **never archive a foundational doc without
honestly updating it first** (owner directive, 2026-06-21).
6
7       ---
8
9       ## 0. How to use this
10      - **Reviewing now?** Read §4 (the open findings/actions) ? that's what needs attention.
§3 is the full register.
11      - **Marking something complete?** Update the doc's own **§7 progress tracker** (the
source of truth for *its* completeness), then flip its row here, then ? if terminal ? run the §2
archival checklist.
12      - **This register does NOT duplicate completeness detail** ? it indexes lifecycle +
```

archive-readiness and points to each tracked doc's §7.

1. Lifecycle legend
`DRAFT ? IN PROGRESS ? DONE/DELIVERED ? ARCHIVED`. Plus: **LIVING** (permanent reference, never "done"), **PROPOSAL/DESIGN** (owner-gated, not started), **REPORT** (a terminal record of a run/launch), **CONVENTION** (a standing rule/guide).

2. Archival due-diligence checklist ? *the repeatable process (do NOT just `git mv`)*
Before a foundational/tracked doc moves to `docs/archives/`:
1. **VERIFY claims against the code** ? no stale assertions; tag load-bearing facts `[verified]`. (A doc that says "nothing built" when code shipped is *not* archive-ready ? fix it first.)
2. **Flip Status ? DELIVERED / DONE / REJECTED + a completion note** ? what shipped, where (PRs, paths, artifacts), and what was **deferred / carried forward**.
3. **Update ALL inbound references** ? `docs/README.md` index, `CLAUDE.md`, `docs/NEXT_STEPS.md`, and any cross-doc links ? the new path + status.
4. **Keep the lineage** ? ADR forward/back-links + "supersedes / superseded-by" so the evolution stays traceable.
5. **`git mv` the whole dir/file ? `docs/archives/past_projects/`** (terminal-state ONLY; live docs stay at `docs/` top level). Relative links inside become historical (note this in the doc's header).
6. **Record the move** ? flip the row here to §3e + a line in memory `current-state`.

> **Worked example (the template):** **DECOUPLED_COMPUTE ? #270** ? verified M0?M2+M3.5 delivered on GCP, flipped ??? DELIVERED with a completion note (incl. carried-forward WASB-C3/DPA/`md5`), reindexed `docs/README` + `past_projects/README`, kept the ADR-010?014 lineage, then `git mv` to `docs/archives/past_projects/DECOUPLED_COMPUTE/`.

3. The register

3a. Tracked projects ? lifecycle docs (archive when DONE; each doc's §7 is the completeness source-of-truth)
Doc	Status	Archive-ready?	Next action
`COMPUTE_DECOUPLED_SERVING/`	**IN PROGRESS** ? M0 approved + **M1?M4 SHIPPED** (#279/#280/#282/#283); M5+ default-OFF code batch authorized (2026-06-21)	No	M5 truly-local ? M6 Cloud Run ? M8 ledger ? M9-auth ? M11 flywheel ? M12 gdrive-api (owner residual: counsel/prices/OAuth/real-run within the D17 cap)
`RALLY_QUALITY_RESEARCH.md`	Foundation, **tracked & ACTIVE** (§7)	No	Drives the R-series; complete when §7.1 single quality number is attributed
`R2_EVAL_METRIC_DESIGN.md`	IMPLEMENTED (augmenting)	No (gates R4 + dual-eval; ablation pending)	Promote-or-reject tolF1 as the gate (ablation) ? then archive with the R-cluster
`R1_MILESTONE_LAUNCH_REPORT.md`	LAUNCHED (#204) ? terminal record	Soon (with the R-cluster)	Keep with R-series for narrative; archive when the track wraps
`RALLY_DETECTION_GAP_CLOSURE.md`	DRAFT (owner-gated)	No	G1?G3 levers; part of the R/quality cluster
`PER_VIDEO_WORKSPACE.md`	DRAFT, IN PROGRESS (P0 = #264)	No	? supersede-reconcile (finding #1) DONE; P3 de-hardcode base ? via #282; remaining P1?P6 (P1 v6 migration coordinates with M8's v6 cost-ledger ? version bump authored once)
`MULTI_SIGNAL_FUSION_PLAN.md`	P1 DONE; P2+ owner-gated	No	P3/P4 (learned fusion, audio) ? needs GPU golden features
`EXPERIMENT_HARNESS.md`	P1 IMPLEMENTED	No	A/B + promotion gate; supports the quality track
`PROMINENT_COURT_DETECTION.md`	DRAFT (owner-gated)	No	C-series; multi-track selection design
`EXECUTION_POLICY.md`	P1 SHIPPED; P3b next	No	Async job policy; reused by Cloud Run dispatch
`GOLDEN_REGRESSION_FIXTURES.md`	IN PROGRESS (owner-gated)	No	Golden eval fixtures
Data layer ? `docs/data_pipeline/` (`GOLDEN_SET_IMPLEMENTATION` · `GOLDEN_SET_PHASE1_VIDEOS` · `GOLDEN_VIDEOS` · `VIDEO_RECORDING_GUIDELINES` · `GOLDEN_DATA_SHARING`)	LIVING (data, **no-ML**)	n/a	Relocated 2026-06-21 to the isolated data pillar; tracked via `data_pipeline/README.md`. Roora *vendor* docs archived ?

`archives/roora-vendor/`. |
`CODE_CLEANUP_AUDIT_2026-06-24.md`	**IN PROGRESS** ? Phase 1: P0/P1/P4/P6a shipped (#352/#355/#354/#356). Quick wins (Q4/Q5/Q6/Q8) next.	No	Q4?Q5?Q6?Q8 ? P6b ? P2 (segmenters) ? P3 (extra=forbid) ? P5 (app-factory) ? P7 verify
`DESKTOP_APP_PLAN.md`	DRAFT (owner-gated, nothing built)	No	P0 compute-feasibility spike; relates to truly-local profile
`OWNED_MODEL_IMPLEMENTATION_PLAN.md`	PLAN/roadmap (owner-gated)	No (authoritative roadmap)	Gen-0?Gen-2; the owned-model track
`PERSONALIZATION_PLAN.md`	DESIGN ? owner-adopted; Phase-0	No	Per-user models (DB v5 store landed)
`UI_ALPHA_EXECUTION_PLAN.md`	ACTIVE	No	khelsutra UI alpha track
`I18N_PLAN.md`	Design (P0); **in implementation** (adapted)	No	Executed in `khelsutra/docs/FRIENDLY_AND_MULTILINGUAL.md` (P1?site/, P3?web/); P2 backend stays here, owner-gated
`POST_166_...RISK_ASSESSMENT.md`	Plan for owner review	No	Milestones/risk doc

3b. Foundational / strategy references (LIVING ? archive only if **superseded**, not when "done")

Doc	Status	Note
`PLATFORM_ARCHITECTURE.md`	Strategy/research	The `ComputeBackend`/`StorageBackend` registries ? realized by COMPUTE_DECOUPLED_SERVING
`COMMERCIALIZATION.md`	? LIVING	Strategy; carries WASB-C3
`VIDEO_LOCALITY_MODEL.md`	THINKING DOC	L0 fully-local = the truly-local profile
`HOSTING_PLAN.md`	PROPOSAL	**? partly overtaken by ADR-013 + the shipped site ? verify (finding #5)**
`UI_PROPOSAL.md`	PROPOSAL (partly shipped)	UX half **in implementation** ? `khelsutra/docs/FRIENDLY_AND_MULTILINGUAL.md`; currency banner added (finding #5 ? for this doc)
`STORAGE_SHARING_MODEL.md`	DRAFT (co-design pending)	?
`ANALYZERS_AND_RECONCILERS.md`	DESIGN (owner-gated)	?
`ROADMAP.md` / `CODE_MAP.md` / `HOW_RALLY_DETECTION_WORKS.md` / `KHELSUTRA_PRODUCT_OVERVIEW.md`	LIVING references	Keep current; not archive targets

3c. Reports & terminal records (archive candidates once their track wraps)

Doc	Status	Note
`REAL_FOOTAGE_VALIDATION_REPORT.md`	validation complete	Motivates R2; archive with the R-cluster
`RALLY_DETECTION_QUALITY_REPORT.md`	report	Quality snapshot; part of R-cluster
`QUALITY_ITERATIONS.md`	LIVING empirical log	**Keep** ? the paper-backbone ablation log ([[paper-coauthor-aim]])
`CODE_AUDIT_AND_TEST_HARDENING.md` · `HARDENING_LOOP_HANDOFF.md` · `DEFERRED_HARDENING_PLAN.md`	COMPLETED ? **top-level pointer stubs** (archived copies exist)	**? confirm intentional stubs (per #165) vs finish-archiving (finding #4)**

3d. Conventions / guidelines / templates / runbooks (permanent ? do not archive)

`PROJECT_DOC_TEMPLATE.md` · `NEXT_STEPS.md` (? refreshed 2026-06-21) · `BACKLOG.md` · `EVALUATION.md` · `ABLATION_LAYER.md` · `SETUP_NEW_MACHINE.md` · **`CREATING_REELS.md`** (? end-user usage guide ? install ? process ? pick ? reel; LIGHT tier; added 2026-06-23) · **`CONVERT_GCS_VIDEOS_TO_HIGHLIGHTS.md`** (? operator runbook ? `gs://` videos ? reels on L4 Cloud Run; verified e2e 2026-06-23) · `TRACKNET_WSL_SETUP.md` · `README.md`. *(2026-06-21 relocation: the data-layer guides moved ? `data_pipeline/`; the field-ops + Roora docs ? `archives/{field-pmf,roora-vendor}/`.)*

3e. Already archived (`docs/archives/`)

`past_projects/DECOUPLED_COMPUTE/` (? 2026-06-21, #270) · `past_projects/low-regret-moves/` · **`field-pmf/`** (field-ops + customer-feedback, 2026-06-21) · **`roora-vendor/`** (annotation-vendor engagement, 2026-06-21) · `PER_VIDEO_OUTPUT_LAYOUT-SUPERSEDED-2026-06-21.md` · `CODE_AUDIT_AND_TEST_HARDENING-COMPLETED-2026-06-14.md` · `DEFERRED_HARDENING_PLAN-COMPLETED-2026-06-14.md` · `HARDENING_LOOP_HANDOFF-COMPLETED-2026-06-14.md` · `COMMERCIAL_READINESS_REVIEW.md` · `MONETIZATION_AUDIT.md` · `TEST_RUN_SUMMARY.md` · `research/` · `checkpoints/` · `quality-iterations/`.

3f. ADR log ? `docs/archives/decisions/DECISIONS.md`

80 **ADR-001 ? ADR-014** (14, all ratified). The compute/serving lineage: **ADR-005 ? 008 ? 009 ? 010 ? 011 ? 012 ? 013 ? 014**. ADR-014 was **RATIFIED 2026-06-21** (owner M0 sign-off for COMPUTE_DECOUPLED_SERVING).
81
82 ## 4. Open due-diligence findings / actions (2026-06-21 review)
83
84 > **? Doc-restructuring done (2026-06-21, owner-directed):** field-ops/customer-feedback ? `archives/field-pmf/`, Roora annotation-vendor ? `archives/roora-vendor/`, and a no-ML **data layer** carved into `docs/data_pipeline/`. Inbound + outbound links rewired; three dir READMEs added (the `data_pipeline` one sets the "no segmentation/stitching" boundary).
85 1. **? RESOLVED (2026-06-21) ? `PER_VIDEO_WORKSPACE.md` supersedes `PER_VIDEO_OUTPUT_LAYOUT.md`.** Folded forward the `video_slug` naming option (workspace §8.5) + the `output/GX010125_run/` manual-prototype precedent; predecessor archived ? `docs/archives/PER_VIDEO_OUTPUT_LAYOUT-SUPERSEDED-2026-06-21.md`; `CODE_MAP.md` repointed.
86 2. **? RESOLVED (2026-06-21) ? `NEXT_STEPS.md` refreshed** with a current 2026-06-21 state + a **cross-track prioritized pending list**; the 2026-06-13 owned-model/quality body is kept (clearly marked) as that track's detail.
87 3. **R-series / quality cluster is ACTIVE** (`R1`,`R2`,`RALLY_DETECTION_GAP_CLOSURE`,`RALLY_QUALITY_RESEARCH`,`QUALITY_ITERATIONS`,`REAL_FOOTAGE_VALIDATION_REPORT`,`RALLY_DETECTION_QUALITY_REPORT`). **Do NOT archive yet** ? archive as a *cluster* once R2 is promoted-or-rejected as the gate + §7.1's single number is attributed. (Owner already confirmed "leave as is", 2026-06-21.)
88 4. **Confirm the COMPLETED top-level stubs** (`CODE_AUDIT_AND_TEST_HARDENING`, `HARDENING_LOOP_HANDOFF`, `DEFERRED_HARDENING_PLAN`) are intentional pointer-stubs (per #165) ? or finish-archive them so there's one copy.
89 5. **`HOSTING_PLAN.md` / `UI_PROPOSAL.md` are PROPOSAL but partly overtaken** by ADR-013 (DO dropped) + the shipped khelsutra site. **? `UI_PROPOSAL.md` banner added (2026-06-21)** ? its UX half is now tracked in `khelsutra/docs/FRIENDLY_AND_MULTILINGUAL.md`; the `I18N_PLAN.md` localization track is likewise in implementation there. **`HOSTING_PLAN.md` banner still pending** (ADR-013 already noted in it).
90 6. **? RESOLVED (2026-06-21) ? `COMPUTE_DECOUPLED_SERVING` M0 approved + M1?M4 SHIPPED** (#279/#280/#282/#283); all §8 gating Qs locked (D7?D17); M5+ default-OFF code batch authorized (owner residual: counsel ToS/DPA, real GCP prices, OAuth app, real cloud run within the D17 $100/?10k cap).
91
92 ## 5. Maintenance
93 Update a row when its doc changes lifecycle; run §2 before any archive; re-review §4 at each clean-slate. This register + the §2 checklist are the operational form of the `CLAUDE.md` archive policy and [[working-agreement-tracked-project-docs]].
94

4. assistant (2026-06-29T17:01:46.267Z)

Let me read the key docs in question and search for inbound references.

5. user (2026-06-29T17:01:47.703Z)

1 <!--
2 PROJECT DOC TEMPLATE ? copy this file to docs/<YOUR_PLAN>.md and fill it in.
3 -->
4
5 # Refactor Rally Detection Edge Cases & Parser Stability
6
7 > **Status:** `DONE` · **Owner:** `Agent` · **Created:** `2026-06-19` · **Last updated:** `2026-06-29`
8 > **Lifecycle:** `DRAFT ? IN PROGRESS ? DONE ? archived` (archival deferred ? see the 2026-06-29 P2b note)
9 > **Tracking anchors:** §7 progress tracker is the source of truth; indexed in `docs/README.md`.
10 > **Relation to existing docs:** Extends `RALLY_DETECTION_QUALITY_REPORT.md`.
11 >
12 > **2026-06-29 update:** the three findings were re-verified against current `master` (post the
13 > 38-commit advance that touched both `rally_gate.py` and `tracknet_runner.py`) ? **all three were

> still present and unfixed.** P0 (the two in-repo edge cases) is **implemented + unit-tested in this
> same PR (#393)**: the `_spread` fix (A) is behaviour-neutral and lands unconditionally; the
> `_trim_inactive_tail` density fix (B) is a behaviour change on sparse footage (R4 is **default-ON**
> in prod) so it lands behind a **default-OFF** `inactivity_temporal_density` flag, promoted only
> after a golden-set measurement (§8 Q2). P1 lives in the sibling `sports-data-collector` repo, so it
> ships as a **separate PR there** (cannot be in #393).
>
> **2026-06-29 ? P2b CLOSED (promoted); plan DONE.** The golden-set A/B ran at the SERVED operating
> point (n=15 clips / 595 rallies; cached WASB trajectories + labels from `gs://khelsutra-rally-corpus`,
> GTs built via the repo's own `load_gt_intervals` so byte-identical to `fusion_golden`'s `gts`):
> **tolF1 0.353 ? 0.364 (?tolF1 +0.010, 95% CI [?0.014,+0.034], sign 10W/4L, p=0.18 ? neutral-to-
> positive, NO regression)**, |?end| 2.97?2.72 s, seg_count_ratio ?0.103 (p=0.006), split_rate ?0.010
> (p=0.016), R1 over-seg guard PASS. `served_regresses(eps=0.0)=False` ?
> `promote_if_served_safe(structural)=promote`. Per the §8 Q2 rule (promote iff tolF1 does not regress)
> the flag was **promoted to default-ON** (`IndexingConfig.inactivity_temporal_density = True`). All
> three findings (A/B/C) are now resolved ? this plan is **DONE**. *Archival to `docs/archives/` is
> deferred until the wider R-series quality cluster is terminal, per `docs/DOC_STATUS.md`.*

---

0. TL;DR
The rally detection pipeline is largely robust, but a deep code review highlighted three minor edge cases/bugs: a potential divide-by-zero or zero-spread in `_spread` for tiny sequences, a density calculation vulnerability for sparse tracks in `_trim_inactive_tail`, and silent skipping of malformed rows in the CVAT parsing script. This plan outlines the fixes for these issues to increase stability.

1. Problem & goal
The current implementation of the candidate windowing handles normal inputs extremely well (including handling blobs), but could fail or behave erratically under specific conditions:
- **Zero-spread calculation**: `_spread` in `rally_gate.py` evaluates to `0.0` for arrays of length 2, which could cause a zero deadband if used on short clips.
- **Sparse trailing frames**: `_trim_inactive_tail` calculates frame density based on *visible* frames. A long gap followed by a single active frame gives 100% density, artificially inflating the rally end.
- **Silent failure on malformed CSV**: `load_canonical_csv` in `rally_cvat.py` silently ignores rows without a valid `rally_number`.

Goal: Fix these three issues to harden the rally detection and ingestion pipelines without changing the existing heuristic thresholds.

4. Design / architecture
The changes are entirely localized to the identified functions:
- `rally_gate.py` ? **done (#393)**: `_spread` now rounds the p95 index UP with `math.ceil()`, so a
  2-element (or any short) input falls back to the full min..max range instead of collapsing to
  `0.0`. On large arrays ceil-vs-truncate differ by at most one sample, so the robust p5/p95 trim is
  unchanged in practice.
- `tracknet_runner.py` ? **done (#393), flag-gated default-OFF**: `_trim_inactive_tail`

gains an `temporal_density` flag. When ON, the density denominator is the trailing window's **temporal length** (`min(trail_frames, frames[i] - frames[0] + 1)`) instead of `win_count` (the count of active *samples*, which sparse tracking inflates). Because `group` holds only active frames, with dense tracking the two denominators are identical ? so flipping the flag is a **no-op on well-tracked clips** and changes only the sparse-tracking case. **Correction (2026-06-29):** R4 is **default-ON in production** (`window_inactivity_end = 1.5`, promoted at the R1 milestone ? an earlier draft of this doc wrongly called it default-OFF), so the fix is a *behaviour change* on sparse footage and ships behind the new `inactivity_temporal_density` flag (**default-OFF**, plumbed config ? `trajectory_hybrid` ? eval `WindowingPreset`), to be promoted only after a golden-set measurement at the SERVED operating point (§8 Q2).
- `rally_cvat.py` ? **separate PR** (sibling `sports-data-collector` repo, not this one): introduce a `logger.warning` before each `continue` in `load_canonical_csv` so skipped/malformed rows are surfaced during ingestion.

6. Honest limits ? what this does NOT do
This will NOT change the core AI validation (Phase 3) or the existing `velocity_thresh` values. It solely tightens up boundary calculation edge cases. It will not suddenly "fix" rallies that the tracking model missed entirely due to lighting or occlusion.

7. Deliverables & progress tracker ? **source of truth**
Legend: ? Todo · ? In progress · ? Done · ? Blocked/gated. **One small PR per row**, branched from `origin/master`, no stacking. Update the row (status + PR link) as work lands.

ID	Deliverable	Depends on	Gated?	Status	PR
P0a	Fix `_spread` zero-collapse (behaviour-neutral, unconditional)	?	No	?	#394
P0b	`_trim_inactive_tail` temporal density behind `inactivity_temporal_density` flag (default-OFF)	?	No	?	#394
P1	Warn on rows silently skipped by `load_canonical_csv` (sibling `sports-data-collector`)	?	No	?	sdc#50
P2a	Expose the flag in the `rally_seg_eval` A/B CLI (`--[vs-]inactivity-temporal-density`) so the promotion run is one command	P0b	No	?	this PR
P2b	**Promote P0b**: golden-set A/B at SERVED windowing ? flip flag default-ON (no regression)	P2a	owner?	?	#396

8. Open questions ? owner / external
1. ~~Do we want `_trim_inactive_tail` to fall back to the existing behavior if there are very few frames, or strictly enforce a minimum temporal density?~~ **Resolved (#393):** strict temporal density. Because `group` carries only active frames, the temporal denominator equals the sample-count denominator under dense tracking, so this is behaviour-preserving for well-tracked clips and only corrects the sparse case ? no separate few-frames fallback needed.
2. **Owner promotion gate (P2b) ? DONE 2026-06-29.** P0b shipped flag-OFF, so the merge was a **zero-regression no-op in production**. The A/B is a single command (P2a wired the toggle into `rally_seg_eval`):

   ```
   ```

```
91	python -m backend.eval.rally_seg_eval \
92	    --manifest <golden_manifest.json> --features-dir <golden_features_dir> \
93	    --preset served --vs served --vs-inactivity-temporal-density
94	```
95
96	(base = SERVED with the flag OFF, candidate = SERVED with it ON; diffs tolF1 at the **SERVED**
97	operating point ? R4 is default-ON in prod, so this measures B's real end-boundary effect on
98	sparse footage.) **Promote** by flipping `IndexingConfig.inactivity_temporal_density` to `True`
99	(a one-line PR) **only if tolF1 does not regress**. This step is owner-only: it needs the golden
100	corpus + GPU/WSL-extracted trajectory features, which don't exist offline. Note finding B is a
101	**no-op under dense tracking**, so the A/B only shows signal on clips where the shuttle tracker
102	actually drops out ? which is exactly why it can't be measured on synthetic/offline fixtures.
103
104	**Measured 2026-06-29 (PR #396).** Ran exactly the command above on the 15-clip golden set
105	(trajectories + labels from `gs://khelsutra-rally-corpus`; GTs via the repo's own
106	`load_gt_intervals`). Result at SERVED: **tolF1 0.353 ? 0.364** (?tolF1 **+0.010**, 95% CI
107	[?0.014, +0.034], sign **10W/4L**, p=0.18 ? neutral-to-positive, **does not regress**); |?end|
108	**2.97 ? 2.72 s**; seg_count_ratio **?0.103** (p=0.006), split_rate **?0.010** (p=0.016); **R1
109	over-seg guard PASS** (net 0.397?0.378). Formal gate: `served_regresses(eps=0.0)=False`,
110	`promote_if_served_safe(structural=True) ? promote`. Per the rule (promote iff tolF1 does not
111	regress) ? **PROMOTED** to default-ON. The stricter honest-LIFT bar reports "no significant lift"
112	(CI spans 0) but **no regression** ? i.e. a do-no-harm correctness fix, not a measured win.
113
114	## 9. Definition of done
115	- [x] `_spread` bug fixed and tested (behaviour-neutral, unconditional).
116	- [x] `_trim_inactive_tail` density fix implemented behind `inactivity_temporal_density` (default-OFF),
117	      plumbed config ? segmenter ? eval preset, tested in both flag states.
118	- [x] **P2 promotion**: golden-set A/B at SERVED windowing, flag flipped default-ON ? no regression
119	      (?tolF1 +0.010, 10W/4L; seg_ratio ?0.103 p=0.006; over-seg guard PASS). *(2026-06-29, PR #396 ? see §8 Q2)*
120	- [x] Warning log added for malformed rows in `rally_cvat.py`. *(sibling-repo PR `sdc#50`)*
121	- [x] Tests updated and passing *(offline suite; golden-corpus regression is owner-gated ? see Q2)*.
122
123	## 10. References
124	**Internal code anchors `[verified]`:**
125	- `backend/pipeline/detectors/rally_gate.py`
126	- `backend/pipeline/detectors/tracknet_runner.py`
127	- `src/parsers/cvat/rally_cvat.py`
128	**Internal docs:**
129	- `docs/RALLY_DETECTION_CODE_REVIEW.md`
130
```

6. user (2026-06-29T17:01:48.940Z)

```
-rw-r--r--  1 avidullu avidullu  6895 Jun 25 11:46 EVALUATION.md
-rw-r--r--  1 avidullu avidullu  4259 Jun 25 11:46 HOW_RALLY_DETECTION_WORKS.md
```

```
-rw-r--r--  1 avidullu avidullu 61174 Jun 25 11:46 QUALITY_ITERATIONS.md
-rw-r--r--  1 avidullu avidullu  6368 Jun 25 11:46 R1_MILESTONE_LAUNCH_REPORT.md
-rw-r--r--  1 avidullu avidullu 14872 Jun 25 11:46 R2_EVAL_METRIC_DESIGN.md
-rw-r--r--  1 avidullu avidullu  6392 Jun 29 10:44 RALLY_DETECTION_CODE_REVIEW.md
-rw-r--r--  1 avidullu avidullu 10020 Jun 29 14:05 RALLY_DETECTION_FIXES_PLAN.md
-rw-r--r--  1 avidullu avidullu 16308 Jun 25 11:46 RALLY_DETECTION_GAP_CLOSURE.md
-rw-r--r--  1 avidullu avidullu 32393 Jun 25 11:46 RALLY_DETECTION_QUALITY_REPORT.md
-rw-r--r--  1 avidullu avidullu 44846 Jun 25 11:46 RALLY_QUALITY_RESEARCH.md
```

7. assistant (2026-06-29T17:01:51.513Z)

Now let me read the R-series cluster docs to assess their terminal/gated status, focusing on their status headers and §7 progress trackers.

8. user (2026-06-29T17:01:56.215Z)

===== RALLY_DETECTION_QUALITY_REPORT.md =====

Local, Cheap, AI Rally Detection for Racket Sports ? Quality Report (Checkpoint)

Internal · honest measured results only · checkpoint 2026-06-17 · supersedes the 2026-06-14 draft
Project: badminton-highlight-indexer · toward a paper

> *An honest empirical study of a shuttle-trajectory rally segmenter, its over-segmentation fix, a
> task-faithful evaluation metric, a disciplined small-N promotion gate, and a first real-footage
> validation against human ground truth and a cloud VLM.*

Abstract

We present a fully local, low-cost rally-detection system for racket sports (badminton first) that
segments a match video into rally `[start, end]` intervals. The system is built on a **frozen
shuttle-trajectory substrate** (a WASB CNN that emits per-frame shuttle position), from which a
velocity signal seeds recall-oriented candidate windows that a tiny pure-numpy logistic-regression
scorer grades. Every additional cue enters as a confidence-weighted **feature column** the scorer
learns to weight ? never a hand-coded branch ? and ships **default-OFF**, promoted only on a measured
held-out lift. We evaluate against an owned golden corpus, now **n=11 labelled matches** (grown from

===== RALLY_DETECTION_CODE_REVIEW.md =====

Rally Detection Code Review

> **2026-06-29 resolution note.** Re-verified against current `master`: all three findings below
> (A `_spread`, B `_trim_inactive_tail`, C `load_canonical_csv`) were still present. **A** is fixed
> unconditionally (behaviour-neutral) and **B** is fixed behind a **default-OFF**
> `inactivity_temporal_density` flag ? both in **PR #393** (this repo). B is flag-gated because R4 is
> **default-ON in production**, so its fix is a real behaviour change on sparse footage that must be
> measured on the golden set before promotion (P2). **C** is tracked for a separate PR in the sibling
> `sports-data-collector` repo. See `docs/RALLY_DETECTION_FIXES_PLAN.md` §7/§8 for status + the
> promotion gate. The analysis below is preserved as the original review snapshot (authored 2026-06-19).

1. How Rally Detection Works

The project identifies rallies through a combination of local trajectory tracking (e.g., TrackNet or WASB) and a series of heuristic gating/trimming rules, culminating in an AI-based refinement (Phase 3). Here's a breakdown of the pipeline:

Phase 2: Candidate Windowing (The Heuristics)

1. **Velocity Thresholding:** The trajectory detector outputs `(Frame, Visibility, X, Y)`. For each visible frame, the normalized velocity (distance divided by frame width per second) is calculated. Frames with a velocity exceeding `velocity_thresh` (e.g., 0.02) are considered "active" (meaning the shuttle is in flight).
2. **Merging / Grouping:** Active frames within a certain dead time (`merge_gap`, usually 2.0s) are merged into candidate windows.
3. **Splitting Over-merged Windows (W1):** If a window exceeds `max_window_duration`, the code splits it at the largest internal gap (dead-time) that is at least `min_split_gap`. If the window lacks gaps (the shuttle never stops moving), a `window_hard_cap` setting (if enabled) forcibly chops it in half at the temporal midpoint.
4. **Inactive Tail Trimming (R4):** Even after a rally finishes, a player might pick up the shuttle and toss it, creating a slow-moving trajectory tail. To prevent the window from overextending, the algorithm trims the end back to the last segment of the window that had a dense cluster of "fast" frames (velocity > `rally_thresh`).

===== QUALITY_ITERATIONS.md =====

Rally-detection quality iterations ? living log

Purpose. A running, reverse-chronological log of how we evolve rally-detection quality: each iteration's hypothesis, what shipped, the measured (or expected) effect, and the lesson. This is the
live continuation of the historical trail in
[`archives/quality-iterations/`](archives/quality-iterations/README.md) ? per the archive policy (`CLAUDE.md`), **terminal-state** (DONE/REJECTED) work gets a dated subdir there; this file stays at
`docs/` top level and keeps growing, then its entries graduate to the archive when they conclude.

Methodology (non-negotiable). Every quality change rides the experiment harness (EXPERIMENT_HARNESS.md): new cues land **flag-gated, default OFF**; they're
graded **offline against golden labels** (per-video + LOVO-honest); promotion to default happens **only
on measured lift** through the `run_regression.py` gate; **rollback = flip a config flag, never `git revert`.** Cues are designed in MULTI_SIGNAL_FUSION_PLAN.md; scoring
in EVALUATION.md; the 2-layer model in
HOW_RALLY_DETECTION_WORKS.md.

Golden corpus = the 15 labelled matches in `output/human_lovo_manifest.json` (the original 6 ?
`adarsh_avi_singles, kushagra_singles, largetest_doubles, gbaaddy, testlarge_short, mahadevpura_singles` ? owner-confirmed 2026-06-13, since grown to 15 with the later GoPro/Boxhill clips). The model trained on these is what we run alongside the existing

===== R2_EVAL_METRIC_DESIGN.md =====

R2 ? Task-faithful rally-boundary eval metric (tolerance-match + over-seg guard)

Status: **IMPLEMENTED (augmenting), 2026-06-17** ? owner-reviewed + decisions locked 2026-06-17;
`backend/eval/tolerance_metrics.py` + `rally_seg_eval` reporting landed (default-additive; tIoU stays the
gate pending the ablation experiment). Gates the rally-end work (R4) and the dual-eval-set regression gate.
A focused increment of the rally-quality project
(RALLY_QUALITY_RESEARCH.md §7;

the day-1 validation that motivates it:
REAL_FOOTAGE_VALIDATION_REPORT.md;
the directive: `eval-boundary-tolerance` + `eval-dual-set-regression`).

> **Implementation refinement (2026-06-17):** §2.3.2's boundary-error distribution is computed over
> **unconditional best-overlap** matches, NOT the within-? 1:1 matches ? a ?-gated set is survivorship-biased
> (loose ends fail the gate and vanish, hiding the very end-deficit the diagnostic exists to surface; that
> within-? deficit instead shows up as low `tol_recall`). Matches the day-1 validation method. Measured on the
> n=9 corpus: **|?start| 1.25 s vs |?end| 2.19 s** ? "start?solved, end=the gap", confirmed.

---

0. TL;DR

`tIoU-F1@0.5` is the wrong ruler for rally detection, in **both** directions: it over-penalizes the
intrinsically fuzzy rally START (the ~1?1.5 s service window), and its overlap matching is murky about

===== RALLY_QUALITY_RESEARCH.md =====

Rally-Detection Quality: Research, Eval Framework & Project Plan

Status: Foundation doc (2026-06-16), run as a **tracked project** (§7) ? *"Bridge the local?Gemini rally-detection gap,"* **completed when all committed tiers land and a single quality
number is attributed (§7.1).** The empirical + research backbone for closing the gap between the
local, no-cloud rally segmenter and the **Gemini** path. Iterations land as their own measured
PRs, logged in [`QUALITY_ITERATIONS.md`](QUALITY_ITERATIONS.md); this doc is the plan they execute against.

How to read this: §1 frames the gap with measured numbers. §2 is the precise, code-grounded
diagnosis (read before proposing fixes). §3 inventories the eval/experiment harness we already
have. §4 specifies how we strengthen that harness (the "really strong evaluation framework").
§5 is the cited external-research synthesis (4 dimensions). §6 is the prioritized roadmap with
falsifiable success criteria. **§7 is the project plan ? work-item tracker, sequencing, the single
number we attribute to the effort, and the definition of done.** §8 recaps the non-negotiable
constraints + cross-links.

> **This doc does NOT duplicate prior design.** It builds on, and cross-links:
> [`HOW_RALLY_DETECTION_WORKS.md`](HOW_RALLY_DETECTION_WORKS.md) (the 2-layer mental model),
> [`EXPERIMENT_HARNESS.md`](EXPERIMENT_HARNESS.md) (the A/B + LOVO harness, P1 shipped),
> [`MULTI_SIGNAL_FUSION_PLAN.md`](MULTI_SIGNAL_FUSION_PLAN.md) (shuttle+person+audio fusion),

===== R1_MILESTONE_LAUNCH_REPORT.md =====

R1 Milestone ? Launch Report (windowing default flip: W1 cap=6 + R4)

Status: LAUNCHED 2026-06-18 (PR #204). **Flip:** the local-windowing defaults go ON ?
`max_window_duration 0?6`, `window_inactivity_end 0?1.5`, `inactivity_rally_thresh 0.08?0.20`,
`inactivity_min_density 0.34?0.30`. **W2 net-crossings, W1.5 hard-cap, and R5 blob-fallback all stay OFF.**
This is the [Launch Report convention](../../) record made **before** the flip merged ? total quality impact
on **both rulers** + the over-seg guard + honest caveats, for posterity.

> **Why this flip is safe to ship:** measured on the **n=13** golden corpus (the grown set, incl. the two
> Boxhill clips GX010137/GX010141), the milestone clears the **net over-seg promotion bar** with room to

> spare, and the verdict was **independently reproduced** (a different harness, #214) and **adversarially
> stress-tested** (5-lens LOO/seed/per-clip workflow). Nothing in the verification refuted it.

---

1. Total quality impact ? BOTH rulers (n=13, leave-one-video-out unit = whole video)

The headline promotion metric is the **R2 task-faithful tolerance-match F1** (`tolF1`, asymmetric ?_start=2/
?_end=1.5); **tIoU-F1@0.5** is kept as the secondary continuity ruler.

===== RALLY_DETECTION_GAP_CLOSURE.md =====

Rally Detection ? Closing the Accuracy Gap (golden-eval-driven, per-video)

> **Status:** `DRAFT` ? implementation **owner-gated** · **Owner:** Avi · **Created:** 2026-06-18 · **Last updated:** 2026-06-18
> **Lifecycle:** `DRAFT ? IN PROGRESS ? DONE ? archived` (move to `docs/archives/` when DONE)
> **Tracking anchors:** §7 progress tracker is the source of truth; pointer in memory `current-state`; add to `docs/README.md` + `docs/NEXT_STEPS.md` when work starts.
> **Relation to existing docs:** **peer-of** [`RALLY_QUALITY_RESEARCH.md`](RALLY_QUALITY_RESEARCH.md) (that is the research + eval-framework doc; this is the *execution tracker* for the remaining accuracy gap). **Builds on** [`MULTI_SIGNAL_FUSION_PLAN.md`](MULTI_SIGNAL_FUSION_PLAN.md) (signal mechanics), [`QUALITY_ITERATIONS.md`](QUALITY_ITERATIONS.md) (the R-log / empirical backbone), [`GOLDEN_VIDEOS.md`](data_pipeline/GOLDEN_VIDEOS.md) (the corpus). Does **not** duplicate them ? it points.
> **Honesty note:** each claim tagged `[verified]` (checked vs code/measured) / `[design]` (proposed). The headline numbers below are real measurements, but `n` is small ? read §6.

---

0. TL;DR

We are using the **growing golden corpus** (human rally labels) to find and close the rally-detection accuracy gap **per video, before launch** ? and the first ground-truth clip already reshaped the picture. The boundary work (R2/R3/R4) **holds** ? when local fires on a real rally, its edges are tight (|?| < 1 s). **The remaining gap is DETECTION, not boundaries:** (G1) **recall** ? both local and Gemini miss ~half the rallies on hard clips; (G2) **precision** ? local over-fires (false positives), especially on continuous drilling; (G3) **rally-END over-extension** ? when a player *quickly knocks the shuttle back after a point*, shuttle-velocity alone can't tell "rally" from "casual reset," so the window runs long / merges. n=2 sharpens the priority: **LOCAL = high recall / low precision, GEMINI = high precision / low recall** (opposite, complementary ? on the normal clip local's recall *beats* Gemini's). So **the #1 lever is precision** (cut local's false positives via `rally_gate` Gate-A + colocation; the player-state rally-END rule feeds this), with **signal fusion** (local's recall + Gemini-style precision) as the durable win. **Caveat:** two golden clips so far (n=15 corpus, GX010141+GX010137) ? directional, not conclusive.

1. Problem & goal

The detector is good enough to ship a highlights product, but the golden eval shows clip-dependent accuracy that we want to **close consistently on every video, not just on average** ? catching the failure modes in the quality loop instead of after launch.

- **Concrete target:** on every corpus video, local is **at-par-or-better than Gemini vs the golden labels**, with **no per-video regression** on either eval set (the dual-eval-set discipline, [[eval-dual-set-regression]]).
- **"Good" =** the three gaps below closed enough that a milestone flip (default-ON) clears the over-seg guard AND improves per-video F1 broadly, recorded in a Launch Report ([[launch-report-convention]]).
- **Read first to get current:** [`RALLY_QUALITY_RESEARCH.md`](RALLY_QUALITY_RESEARCH.md) §1?2 (the gap + diagnosis), [`MULTI_SIGNAL_FUSION_PLAN.md`](MULTI_SIGNAL_FUSION_PLAN.md) §4 (in-flight vs in-hand), [`HOW_RALLY_DETECTION_WORKS.md`](HOW_RALLY_DETECTION_WORKS.md) (the 2-layer

model).

===== REAL_FOOTAGE_VALIDATION_REPORT.md =====

Local (No-AI) Rally Detector ? Real-Footage Validation & Improvement Roadmap

Date: 2026-06-16 · **Audience:** owner/cofounder · **Status:** validation complete, roadmap proposed (all flags default-OFF)
Footage: 6 Kushagra/Yash June-12 clips, 1080p proxies, fully-local no-AI WASB windowing vs Gemini (strong reference, *not* ground truth except mahadevpura-2, owner hand-labeled).
Tracked doc to fold this into: `docs/GOLDEN_REGRESSION_FIXTURES.md` · root-cause backbone: `docs/RALLY_QUALITY_RESEARCH.md` §2.

---

1. Headline verdict

The W1 `max_window_duration` cap **fixed the over-merge catastrophe** ? aggregate count went from 48 (29% of Gemini's 165) to 250 (152%), and mIoU rose 0.23 ? 0.39 ? but it **overshot into over-segmentation and never landed near 1.0×**, so we traded a merge problem for a split problem rather than solving localization. On the one clip with owner ground truth (mahadevpura-2), the best local config is **~86% of Gemini at loose tolerance (F1@0.25 0.722 vs 0.835) but only ~53% at tight boundaries (F1@0.5 0.296 vs 0.557)** ? and the entire residual gap is **rally-END over-extension** (|?end| 3.12s vs Gemini 1.19s), while rally-START is already near Gemini parity (|?start| 1.74s vs 1.46s).

---

2. What worked ? the over-merge fix at scale

The baseline collapsed entire games into mega-windows. The cap demolished that pathology:

Metric (aggregate, n=6)	Baseline	W1 (cap=12)

===== HOW_RALLY_DETECTION_WORKS.md =====

How rally detection works (the 2-layer mental model)

A short conceptual map of how we go from raw video to rally `[start, end]` segments ? and, crucially, **which layer our annotation can improve.**

```
  video ??? frames ??? [DETECTION] ??? shuttle (x,y,visible) per frame
                           ?                    ?
                      (WASB CNN,          per-frame trajectory
                       frozen weights)         ?
                                               ?
                                         velocity signal  (how fast the shuttle moves)
                                               ?
                                         [SEGMENTATION]  ??? the 3 knobs we tune
                                               ?
                                               ?
                                      rally windows  [start, end] ??? IoU vs golden labels
```

Two layers ? keep them separate in your head

9. assistant (2026-06-29T17:01:59.596Z)

Now let me look at the §7 progress trackers and gate status for the key cluster docs (RALLY_QUALITY_RESEARCH §7.1 single quality number, R2 ablation status). Let me also check the R2 doc more fully for the tolfF1 gate/ablation question.

10. user (2026-06-29T17:02:03.891Z)

```
########## R2_EVAL_METRIC_DESIGN.md ? ablation/gate section ##########
4:`backend/eval/tolerance_metrics.py` + `rally_seg_eval` reporting landed (default-additive;
tIoU stays the
5:gate pending the ablation experiment). Gates the rally-end work (R4) and the dual-eval-set
regression gate.
12:> (loose ends fail the gate and vanish, hiding the very end-deficit the diagnostic exists to
surface; that
86:5. **`tIoU`/`mIoU` demoted to a SECONDARY trend-line diagnostic** (keeps the paper-ablation
backbone
87:   comparable) ? **not the gate.**
90:The preview (§3) shows even **Gemini** scores `tolF1@1.5 = 0.20` on mp2 ? i.e. on ~half the
rallies the
104:| clip / method | tolF1@1.5 | @2.0 | @2.5 | @3.0 | (tIoU-F1@0.5) |
112:- **`tolF1@~3.0 ? tIoU-F1@0.5`** ? the sweep is a smooth, interpretable knob that *contains*
tIoU as a point;
156:- **Default-additive ? ablation-gated replacement:** R2 is reported next to tIoU (augment).
The *gate*
157:  switches to R2 only after a dedicated **ablation experiment** on the golden corpus shows
R2 ranks candidates
159:  as the gate it is **deprecated with documentation, not deleted** (kept as the secondary
trend-line + the
160:  `QUALITY_ITERATIONS.md` paper-ablation backbone, with a dated note recording the ablation
evidence for the
165:## 6. Definition of done
166:- `tolerance_metrics.py` + tests merged (default-additive); `rally_seg_eval` reports the R2
decomposition
171:  until the ablation experiment (§7) validates it as the gate; only then does the gate
switch.
172:- tIoU/mIoU retained as a secondary trend-line; when retired as the gate, **deprecated-with-
docs (kept for
173:  posterity)** per the §7 owner decision ? a dated note records why and links the ablation
evidence.
179:never one number; tIoU demoted to secondary; R2 default-additive until validated; boundary-
error split
188:- **Gate: AUGMENT now, then ablation-gate the eventual replacement, then deprecate-with-docs
? ADOPTED.**
190:  R2 becomes the promotion gate, run a dedicated **ablation experiment** on the golden
corpus showing R2
192:  retired as the gate, it is **formally deprecated with documentation ? kept for posterity,
never deleted**:
193:  it remains a secondary trend-line and the paper-ablation backbone
(`QUALITY_ITERATIONS.md`), with a dated
194:  note recording *why* it was superseded and the ablation evidence behind the switch.
(Consistent with the

########## RALLY_QUALITY_RESEARCH.md ? §7 / single number / DoD ##########
3:**Status:** Foundation doc (2026-06-16), run as a **tracked project** (§7) ? *"Bridge the
5:number is attributed (§7.1).** The empirical + research backbone for closing the gap between
the
14:number we attribute to the effort, and the definition of done.** §8 recaps the non-negotiable
286:- **CausalTAD** (2024, [arXiv:2407.17792](https://arxiv.org/abs/2407.17792)) ? *steering-
away note.*
499:## 7. Project: plan, tracking & definition of done
502:**completes when all committed tiers land AND a single quality number is attributed to the
effort**,
503:measured honestly and logged. This §7 is the living burn-down (update the status column as
PRs merge;
507:### 7.1 The number (what we attribute to this effort)
532:### 7.2 Work items (status checklist)
533:Status: ? not started · ? in progress · ? done. (Each row = its own default-OFF, measured
PR.)
535:| ID | Deliverable | Tier | Depends on | Acceptance / promote criterion | Status |
553:### 7.4 Definition of done
```

557:2. **The headline number (§7.1) is recorded** in `QUALITY_ITERATIONS.md` + the leaderboard, with honest
564:> overfitting at n=6?16. **Default:** the project *completes* at Tier 0?2 + an attributed number, and

11. assistant (2026-06-29T17:02:07.073Z)

Let me read the key gate-determining sections: R2 §6/§7 ablation status, and RALLY_QUALITY_RESEARCH §7.1/§7.2/§7.4 to confirm whether the cluster gate is closed.

12. user (2026-06-29T17:02:08.948Z)

499   ## 7. Project: plan, tracking & definition of done
500
501   This roadmap is run as **one tracked project ? "Bridge the local?Gemini rally-detection gap."** It
502   **completes when all committed tiers land AND a single quality number is attributed to the effort**,
503   measured honestly and logged. This §7 is the living burn-down (update the status column as PRs merge;
504   log each result in [`QUALITY_ITERATIONS.md`](QUALITY_ITERATIONS.md), numbers in
505   `eval_baselines/experiment_leaderboard.json`).
506
507   ### 7.1 The number (what we attribute to this effort)
508   Measured **leave-one-video-out, grouped by match, on the golden set** (grown to ?~16):
509   - **Headline:** multi-court rally-segmentation **tIoU-F1**, `baseline ? final`, plus **gap-closure %**
510   = `(final ? baseline) / (Gemini ? baseline)`. Today: heuristic **0.157** ? Gemini **0.66**
511   (Gen-0 head already **0.561**). **Success bar (proposed, owner-adjustable): close ? 50% of the
512   heuristic?Gemini multi-court gap** ? i.e. local multi-court F1 ? ~0.41, ideally beating Gen-0's 0.561.
513   - **Primary lens (over-segmentation-sensitive, the actual failure):** segmental **F1@0.25** +
514   **Edit-score** + **merge-rate** ? these must improve even if tIoU-F1 moves little.
515   - **Reported with honest stats** (bootstrap CI + paired sign-test); a result counts only if it also
516   holds on the **served** windowing (eval?serve gate). The **baseline is frozen at E1** before any fix.
517   - **Tracked headline = the n=6 golden *aggregate* tIoU-F1@0.5** (LOVO; the robust unit, not a single
518   video) via `python -m backend.eval.rally_seg_eval`. The 0.157/0.66 figures above are per-venue
519   *references*; computing an exact "% gap-to-Gemini closed" needs Gemini measured on the *same* golden
520   set + metric ? **TODO** (don't mix the per-venue refs with the aggregate).
521
522   **Progress (cumulative, golden n=6 aggregate tIoU-F1@0.5):**
523
524 | after | F1 | merge_rate | ? vs baseline |
525 |---|---|---|---|
526 | E1 baseline (current heuristic) | 0.300 | 0.523 | ? |
527 | + W1 bounded windowing (cap=12, **default-OFF**) | **0.439** | **0.038** | **+0.139** (6/0 sign-test) |
528 | + W2 net-crossings gate (mc=1, **default-OFF**) | **0.457** | 0.038 | **+0.157 cumulative** (W2 step +0.019, 6/0) |
529 | + W1.5 hard-cap fallback (**default-OFF**) | 0.444 | 0.000 | **+0.006 (n.s., CI spans 0)** ? neutral on golden; real-footage fix (mahadevpura-1 174s blob ? 24×?9s windows) |
530 | + W2 multi-feature scorer (person/audio) | _next (GPU)_ | | |
531
532   ### 7.2 Work items (status checklist)
533   Status: ? not started · ? in progress · ? done. (Each row = its own default-OFF, measured PR.)
534

ID	Deliverable	Tier	Depends on	Acceptance / promote criterion	Status
E1	merge/split-rate metric + F1@k surfaced + one-command `rally_seg_eval` entrypoint; **froze baseline** (#185)	0	?	over-merge shows up as a number on the 6 golden videos	? baseline F1 0.300, merge_rate 0.523
W1	Bounded windowing ? `max_window_duration` cap + inactivity split (See et al.) (#187)	1	E1	seg F1@k ? vs baseline (CI?0 **and** sign-test), no served regression; merge-rate ?	? **?F1 +0.139 (6/0), merge_rate 0.523?0.038** (default-OFF)
W1.5	Hard-cap fallback (`window_hard_cap`) ? chop continuously-active windows at the cap when no inactivity gap exists	1	W1	golden ?F1 ?0 (or neutral) **+** fixes a real-footage over-merge it can't otherwise cap	? **golden ?F1 +0.006 (n.s.)**; real `mahadevpura-1` 174s blob ? 24×?9s; retained **default-OFF** (real-footage + true-cap rationale, not a golden lift)
W2	Rally-state features (net-crossings ? #188 + **never-empty safeguard** ?; two-sided motion, **direction-reversal/peak**, audio = follow-on)	1	E1, W1	?1 cue ?F1 CI clears 0 + sign-test, survives served re-check ? promote; others retained default-OFF	? **net-crossings ?F1 +0.019 (6/0)**, gate now **do-no-harm** (never zeroes a video); multi-feature scorer next (GPU)
W3	Keep scorer parameter-light (standing decision; TemporalMaxer evidence)	1	?	documented; no learned temporal net until corpus ? now	?
M1	Serve-start classifier ? rally-start boundary (pose/court)	2	W1, **court-mask decision (owner-gated)**	start-boundary timing error ?; promote on seg-F1 lift, no served regression	?
M2	Per-venue threshold calibration vs golden	2	E1	multi-court F1 ? toward Gen-0 0.561 without harming clean-court	?
R1	Owned segmenter with an explicit **boundary head** (port ASRF/TriDet *idea*, trained ? not grafted)	3 (stretch)	corpus ?~16?30, W1/W2	beats tuned heuristic+LogReg on held-out matches (seg Edit + F1@k), LOVO	?
R2	Timestamp/sparse-supervision pilot (label-efficiency)	3 (stretch)	R1 path	measured label-efficiency win	?
C	Golden corpus 6 ? ~16 (Roora) + active-learning selection + measure IAA	ongoing	?	corpus ?16; IAA recorded	? (~6 now; ~10 by the weekend)

7.3 Sequencing
`E1` (metrics + baseline) ? `W1`/`W2`/`W3` (the high-confidence wins) ? `M1`/`M2` ? `R1`/`R2` (corpus-gated). `C` (corpus growth) runs in parallel throughout ? it is the single highest-leverage input and gates the Tier-3 bets.

7.4 Definition of done
The project is **complete** when:
1. **Tier 0 + Tier 1 + Tier 2** work items are merged ? each a measured PR (default-OFF, promoted only on measured lift, re-confirmed on the served windowing).
2. **The headline number (§7.1) is recorded** in `QUALITY_ITERATIONS.md` + the leaderboard, with honest stats, on the grown golden set ? **and the gap-closure target is met, or the honest null is recorded with a reasoned go/no-go on Tier 3.**
3. Each landed item left the harness greener (no regression) and `QUALITY_ITERATIONS.md` tells the story.

> **Tier-3 scope decision (owner-adjustable):** R1/R2 (the owned boundary-head model + sparse supervision) are **corpus-gated stretch goals** ? the verified research says deep TAL detectors risk overfitting at n=6?16. **Default:** the project *completes* at Tier 0?2 + an attributed number, and Tier 3 spins out into the owned-model milestone ([`OWNED_MODEL_IMPLEMENTATION_PLAN.md`](OWNED_MODEL_IMPLEMENTATION_PLAN.md)).
> Fold Tier 3 into the DoD instead if you want the project to span the owned model.

```
## 8. Constraints & guardrails (non-negotiable) + cross-links

- **Licensing:** every dependency ? code, data, AND weights ? must be **MIT / Apache-2.0 / BSD**.
  Reject viral/NC/custom (Gemma-3, Llama-4 MAU cap, Commons-Clause).
```

13. user (2026-06-29T17:02:09.800Z)

```
- **`run_regression.py`** ? add the dual-set R2 aggregate + the over-seg-guard gate. *(Status: the guard
  primitive landed as `tolerance_metrics.over_seg_guard` and is wired into `rally_seg_eval.compare()`/`--vs`
  ? the natural home, since `rally_seg_eval` loads the golden manifest + trajectories. The dual-set
  `run_regression.py` aggregate is still a separate future item.)*
- **Tests** ? synthetic intervals: exact-match, start-fuzzy-within-?, end-loose-beyond-?, over-production
  (precision crash), the Hungarian-vs-greedy tie case, empty inputs.
- **Default-additive ? ablation-gated replacement:** R2 is reported next to tIoU (augment). The *gate*
  switches to R2 only after a dedicated **ablation experiment** on the golden corpus shows R2 ranks candidates
  at least as sensibly as tIoU ? so the promotion bar never changes silently mid-flight. When tIoU is retired
  as the gate it is **deprecated with documentation, not deleted** (kept as the secondary trend-line + the
  `QUALITY_ITERATIONS.md` paper-ablation backbone, with a dated note recording the ablation evidence for the
  switch). Owner decision, 2026-06-17 (§7).

---

## 6. Definition of done
- `tolerance_metrics.py` + tests merged (default-additive); `rally_seg_eval` reports the R2 decomposition
  next to tIoU (the ?-sweep helper exists in `tolerance_metrics` but is not yet wired into the
  `rally_seg_eval` report); the corpus (n?8) re-scored under R2 and recorded in QUALITY_ITERATIONS.
- The IAA study (§2.4) scoped (it can land after R2's code; R2 ships with provisional ? + the sweep).
- `run_regression.py` carries the dual-set R2 gate + the per-video over-seg guard ? but R2 **augments** tIoU
  until the ablation experiment (§7) validates it as the gate; only then does the gate switch.
- tIoU/mIoU retained as a secondary trend-line; when retired as the gate, **deprecated-with-docs (kept for
  posterity)** per the §7 owner decision ? a dated note records why and links the ablation evidence.

---

## 7. Decisions
**Locked (recommended, design-level):** Hungarian 1:1 (not greedy/overlap); report the decomposition + sweep,
never one number; tIoU demoted to secondary; R2 default-additive until validated; boundary-error split
start/end is the R4-aiming diagnostic.

**Locked (owner review, 2026-06-17):**
- **?_end asymmetric (1.5 s) vs ?_start (2.0 s) ? ADOPTED.** Asymmetry is the point: forgive the intrinsically
  fuzzy start, keep the rally-end honest. Your "±2 s, not more" bar maps to `?_start=2.0`; `?_end` stays the
  tighter, IAA-calibrated knob (§2.4).
```

- **IAA study after R2 code ? ADOPTED.** R2 ships now with provisional `?_start=2.0 / ?_end=1.5` + the full
  sweep; the IAA study lands next to fix ?. R2's code is not blocked on it.
- **Gate: AUGMENT now, then ablation-gate the eventual replacement, then deprecate-with-docs ? ADOPTED.**
  R2 is reported **alongside** tIoU (augment) until validated. tIoU is **not** swapped out silently: before
  R2 becomes the promotion gate, run a dedicated **ablation experiment** on the golden corpus showing R2
  ranks candidates at least as sensibly as tIoU (and exposes the deficits tIoU hid). When tIoU is finally
  retired as the gate, it is **formally deprecated with documentation ? kept for posterity, never deleted**:
  it remains a secondary trend-line and the paper-ablation backbone (`QUALITY_ITERATIONS.md`), with a dated
  note recording *why* it was superseded and the ablation evidence behind the switch. (Consistent with the
  repo's "retain weak/superseded signals; numbers inform, foundations decide" norm.)

14. assistant (2026-06-29T17:02:14.636Z)

I now have a clear picture of the cluster gate. The key insight: P2b promoted the `inactivity_temporal_density` flag (a finding-B do-no-harm correctness fix, NO measured lift), but the DOC_STATUS §4 finding #3 gate is two-pronged: (1) **R2 tolF1 promoted-or-rejected as the gate via the ablation experiment**, and (2) **§7.1 single quality number attributed**. Let me verify P2b did NOT close either prong, and gather all inbound references.

15. user (2026-06-29T17:02:20.043Z)

```
================ INBOUND: RALLY_DETECTION_FIXES_PLAN.md ================
docs/RALLY_DETECTION_CODE_REVIEW.md:9:RALLY_DETECTION_FIXES_PLAN.md
docs/NEXT_STEPS.md:25:RALLY_DETECTION_FIXES_PLAN.md

================ INBOUND: RALLY_DETECTION_FIXES_PLAN (no .md) ================
```

16. user (2026-06-29T17:02:22.116Z)

```
=== exists? NEXT_STEPS.md, README.md ===
ls: cannot access 'NEXT_STEPS.md': No such file or directory
docs/NEXT_STEPS.md
docs/README.md

=== R2 ablation experiment status grep across docs ===
docs/NEXT_STEPS.md:21:8. **[owner-side GPU] R-series: promote-or-reject `R2` tolF1 as the gate** (the ablation) ? needs GPU-extracted golden features.
docs/QUALITY_ITERATIONS.md:186:`rally_seg_eval.compare()` / `--vs`). R2 §2.3.3's *strict* rule blocks promotion if the candidate raises
```

17. assistant (2026-06-29T17:02:25.870Z)

Key confirmation: NEXT_STEPS.md line 21 still lists "promote-or-reject R2 tolF1 as the gate (the ablation)" as a pending owner-side GPU item. The cluster gate is NOT closed. Now let me get the exact inbound references for FIXES_PLAN (in README.md too) and check the R2/cluster references. Let me look at the surrounding context of the two FIXES_PLAN references and check README.md for the FIXES_PLAN index entry.

18. user (2026-06-29T17:02:29.233Z)

```
=== README.md any rally/fixes refs ===

=== context: NEXT_STEPS.md:25 (the FIXES_PLAN inbound) ===
7. **[owner] Set the ship-gate target (#172)** ? needs corpus growth.
8. **[owner-side GPU] R-series: promote-or-reject `R2` tolF1 as the gate** (the ablation) ?
```

needs GPU-extracted golden features.

P2 ? later / gated / deferred
9. Rally-quality **P3/P4** (person-fusion LOVO, audio cue) ? needs GPU golden features.
9b. **[owner · GPU golden] Promote the R4 density fix (finding B / P2 in `RALLY_DETECTION_FIXES_PLAN.md`).** PR #393 landed it behind `inactivity_temporal_density` (**default-OFF**, zero-regression no-op in prod). To ship: golden-set A/B with the flag ON vs OFF at the **SERVED** windowing (the eval `WindowingPreset` carries it); flip the config default to True in a one-line PR only if tolF1 doesn't regress. Fixes a sparse-tracking quirk where one fast sample after a detection gap held the rally tail open (R4 is default-ON in prod).
10. **M0.4 scorer swap** ? TemporalMaxer (per the 2026-06-17 external review).
11. **Multisport** (table-tennis first, then pickleball).
12. Serving **M8/M9** ? per-video cost ledger, edge auth, DPA/consent (within `COMPUTE_DECOUPLED_SERVING`; owner/legal-gated).

=== context: RALLY_DETECTION_CODE_REVIEW.md:9 (already shown header above; confirm it's the resolution note) ===
> measured on the golden set before promotion (P2). **C** is tracked for a separate PR in the sibling
> `sports-data-collector` repo. See `docs/RALLY_DETECTION_FIXES_PLAN.md` §7/§8 for status + the
> promotion gate. The analysis below is preserved as the original review snapshot (authored 2026-06-19).

19. assistant (2026-06-29T17:02:33.797Z)

The grep showed the README.md line for FIXES_PLAN didn't appear ? but DOC_STATUS §9 of FIXES_PLAN's own header says it's "indexed in docs/README.md". Let me verify whether README actually indexes it (the grep with -o may have been line-based; let me grep more broadly) and confirm NEXT_STEPS line 25 references FIXES_PLAN. Also note NEXT_STEPS line 25 (item 9b) references FIXES_PLAN but is now STALE (says P2/finding B not yet promoted, default-OFF ? but #396 promoted it default-ON).

20. user (2026-06-29T17:02:41.415Z)

```
=== Full filename grep in README.md (case-insensitive, partial) ===
  (none found in README.md)

=== Where exactly is FIXES_PLAN.md referenced (line numbers) ===
docs/RALLY_DETECTION_CODE_REVIEW.md:9:> `sports-data-collector` repo. See
`docs/RALLY_DETECTION_FIXES_PLAN.md` §7/§8 for status + the
docs/NEXT_STEPS.md:25:9b. **[owner · GPU golden] Promote the R4 density fix (finding B / P2 in
`RALLY_DETECTION_FIXES_PLAN.md`).** PR #393 landed it behind `inactivity_temporal_density`
(**default-OFF**, zero-regression no-op in prod). To ship: golden-set A/B with the flag ON vs
OFF at the **SERVED** windowing (the eval `WindowingPreset` carries it); flip the config default
to True in a one-line PR only if tolF1 doesn't regress. Fixes a sparse-tracking quirk where one
fast sample after a detection gap held the rally tail open (R4 is default-ON in prod).

=== Does DOC_STATUS.md itself reference FIXES_PLAN or the R-cluster docs by name? ===
34:| `RALLY_QUALITY_RESEARCH.md` | Foundation, **tracked & ACTIVE** (§7) | No | Drives the
R-series; complete when §7.1 single quality number is attributed |
35:| `R2_EVAL_METRIC_DESIGN.md` | IMPLEMENTED (augmenting) | No (gates R4 + dual-eval; ablation
pending) | Promote-or-reject tolF1 as the gate (ablation) ? then archive with the R-cluster |
36:| `R1_MILESTONE_LAUNCH_REPORT.md` | LAUNCHED (#204) ? terminal record | Soon (with the
R-cluster) | Keep with R-series for narrative; archive when the track wraps |
37:| `RALLY_DETECTION_GAP_CLOSURE.md` | DRAFT (owner-gated) | No | G1?G3 levers; part of the
R/quality cluster |
68:| `REAL_FOOTAGE_VALIDATION_REPORT.md` | validation complete | Motivates R2; archive with the
R-cluster |
69:| `RALLY_DETECTION_QUALITY_REPORT.md` | report | Quality snapshot; part of R-cluster |
70:| `QUALITY_ITERATIONS.md` | LIVING empirical log | **Keep** ? the paper-backbone ablation log
([[paper-coauthor-aim]]) |
87:3. **R-series / quality cluster is ACTIVE** (`R1`,`R2`,`RALLY_DETECTION_GAP_CLOSURE`,`RALLY_Q
UALITY_RESEARCH`,`QUALITY_ITERATIONS`,`REAL_FOOTAGE_VALIDATION_REPORT`,`RALLY_DETECTION_QUALITY_
REPORT`). **Do NOT archive yet** ? archive as a *cluster* once R2 is promoted-or-rejected as the
gate + §7.1's single number is attributed. (Owner already confirmed "leave as is", 2026-06-21.)
```

21. assistant (2026-06-29T17:02:45.553Z)

I notice DOC_STATUS.md does NOT track RALLY_DETECTION_FIXES_PLAN.md or RALLY_DETECTION_CODE_REVIEW.md in its register at all. The FIXES_PLAN is a NEW DONE doc not yet registered. Let me confirm the README.md index situation (FIXES_PLAN claims it's indexed there but isn't) and check the FIXES_PLAN's own header lineage cross-ref to QUALITY_REPORT/CODE_REVIEW. Let me also verify whether FIXES_PLAN/CODE_REVIEW are part of the gated R-cluster or standalone.

22. user (2026-06-29T17:02:55.768Z)

```
=== README.md: how does it index rally docs (to see if FIXES_PLAN/CODE_REVIEW belong) ===
1:- [RALLY_QUALITY_RESEARCH.md](RALLY_QUALITY_RESEARCH.md) ? **? Foundation doc** for closing
the local-vs-Gemini rally-detection gap: the measured gap + the code-grounded over-merge
diagnosis ("shuttle moving ? rally"), an inventory of the eval/experiment harness + its gaps,
the strengthened evaluation framework (segmental Edit/F1@k that make over-segmentation first-
class, golden-corpus + active learning, Gemini-agreement as a *shadow* signal), an
**adversarially-verified, cited external-research synthesis** (TAL/TAS boundary heads, racket-
sports rally-state cues, small-data, eval rigor), a **prioritized cheap?durable roadmap with
falsifiable success criteria**, and (§7) a **tracked project plan** ? work-item checklist,
sequencing, the single quality **number** attributed to the effort, and a **definition of done**
(project completes when all tiers land + the number is recorded). The plan that the
[QUALITY_ITERATIONS.md](QUALITY_ITERATIONS.md) log executes against; ties together
EXPERIMENT_HARNESS / MULTI_SIGNAL_FUSION_PLAN / ANALYZERS_AND_RECONCILERS /
OWNED_MODEL_IMPLEMENTATION_PLAN.
2:- [RALLY_DETECTION_GAP_CLOSURE.md](RALLY_DETECTION_GAP_CLOSURE.md) ? **Tracked execution doc
(2026-06-18)** for closing the remaining rally-detection accuracy gap **per video, before
launch**, driven by the growing golden corpus. First ground-truth clip (GX010141) reframed it:
boundaries are tight (R2/R3/R4 holds) ? the gap is **detection**: (G1) recall (local & Gemini
both miss ~half on hard clips, and *different* rallies ? complementary), (G2) precision (local
over-fires), (G3) **rally-END over-extension on the quick-return-after-point** (shuttle still
moving ? needs a *player-state* signal). Levers (owner-gated): player-kinematics + shuttle-in-
hand / sustained-invisibility end-rule with reverse-timeline backtrack, `rally_gate` Gate A
wiring, complementary fusion. **Peer** of `RALLY_QUALITY_RESEARCH.md` (execution tracker, not
research); §7 is the source of truth.
3:- [PROMINENT_COURT_DETECTION.md](PROMINENT_COURT_DETECTION.md) ? **Design + experiment
(2026-06-19, default-OFF, owner-gated flip)** for the **multicourt G2 over-fire**: identify the
**prominent (foreground) court** (the one covering the majority of the frame) and keep only
rallies whose shuttle lives inside it ? re-defining the shuttle signal `(x,y,visible)` ?
`(x,y,visible-AND-inside-prominent-court)` *before* windowing, so a background/adjacent-court
shuttle never forms a rally. 2D floor polygon (MVP) ? **pseudo-3D play-volume** (refinement, so
high clears aren't clipped). Born from the GX010142 rally-#1 background-court FP; reuses
`fusion_features.point_in_polygon` + `FusionConfig.court_polygon`. §8 progress tracker is the
source of truth.
4:- [RALLY_DETECTION_QUALITY_REPORT.md](RALLY_DETECTION_QUALITY_REPORT.md) ? **Technical-paper
checkpoint (2026-06-17)** of rally-detection quality: the honest measured state (over-
segmentation milestone tolerance-F1 0.091?0.323 at n=11, p=0.002; merge-rate 0.694?0.150; first
real-footage ground-truth at-par with START at Gemini parity), the task-faithful R2 metric, a
detailed Gemini comparison, next steps + expected wins, areas yet to be explored, and a
publishability / path-to-submission verdict. Renders to a polished PDF; the externally-shareable
"toward a paper" summary of `RALLY_QUALITY_RESEARCH.md` + `QUALITY_ITERATIONS.md`.
5:- [EVALUATION.md](EVALUATION.md) ? How we measure and improve quality.
6:- [EXPERIMENT_HARNESS.md](EXPERIMENT_HARNESS.md) ? Design (P1 SHIPPED) for A/B + shadow + SxS
experimentation on rally-detection variants with **zero production regression** ? experiments
are decoupled from deploy (new cues default OFF, promote only on measured LOVO lift, rollback =
flip a flag). Composes the existing `backend/eval/` machinery (`calibration`, `regression`,
`metrics`); implemented in `backend/eval/experiment.py`.
7:- [QUALITY_ITERATIONS.md](QUALITY_ITERATIONS.md) ? **Living log** of rally-detection quality
iterations (hypothesis ? what shipped ? measured/expected effect ? lesson). The reverse-
chronological record; terminal-state snapshots get archived under `archives/quality-
iterations/`.
8:- [ANALYZERS_AND_RECONCILERS.md](ANALYZERS_AND_RECONCILERS.md) ? Design (owner-gated, NOT
started): a **signal-fusion framework** for rally detection ? producers emit confidence-scored
**signals** at three temporal **scopes** (frame detectors / window analyzers / session
analyzers), a learned **scorer** weights them, and an auditable **report-first reconciliation**
```

pass lints the decision against the evidence. The spine: **no special-casing in the decision path** (signals?scorer, never `if/else`). Worked examples: a service-fault analyzer, a warm-up/practice span detector, and consuming a per-frame shuttle-state detector. A peer to `EXPERIMENT_HARNESS.md` and `MULTI_SIGNAL_FUSION_PLAN.md` (ADR-007 modular rally layer).
11:- GOLDEN_REGRESSION_FIXTURES.md ? **Design (owner-gated, NOT started):** video-free, non-attributable regression fixtures so a clean clone runs the rally-seg suite with **no video/GPU/DB**. Two layers ? the post-detector freeze + **dual gates** (characterization + quality-floor) reusing the `backend/eval/` stack, and the **SEALED-FIXTURES** privacy filter (de-attribution-at-source, tiered public/key-gated, shuttle-only public tier). Carries a cited **IN/US/EU legal memo** (derived-data IP/privacy), a threat model, and a **§7 progress tracker + definition-of-done**. Extends [GOLDEN_DATA_SHARING.md](data_pipeline/GOLDEN_DATA_SHARING.md).
18:- [DECOUPLED_COMPUTE/](archives/past_projects/DECOUPLED_COMPUTE/README.md) ? ? **DELIVERED + ARCHIVED (2026-06-21).** Lifted the rally pipeline onto a **rented cloud-native GPU** doing bucket-driven `train` + `infer`: storage/compute seams (PRs #246?#259) + the **M3.5 native WASB runner**; **M0?M2 + M3.5 proven on a GCP L4** (M2 ? `weights/0.1.0-l4/`). Delivered on **GCP** ? DigitalOcean dropped for compute ([ADR-013](archives/decisions/DECISIONS.md)). **M3/M4 (serving endpoint + per-video billing + scale-to-zero) deferred ? the "Cloud Run + GPU serving" subproject** ([ADR-012](archives/decisions/DECISIONS.md)). Snapshot under `docs/archives/past_projects/DECOUPLED_COMPUTE/` (incl. `DEPLOY_REPORT_GCP_2026-06-20.md`).
26:- DATA_IN_GCS.md ? **? The corpus source-of-truth map: where every piece of data lives in `gs://khelsutra-rally-corpus`** (canonical layout · current contents + drift · how `backend.worker` consumes it · the still-local-only **gap** · migration plan). Read this so a session pulls data from the bucket, not from someone's `C:`/`F:`/Drive ? the goal is to remove the local box as a blocker. Pairs with the two runbooks below.
35:- PROJECT_DOC_TEMPLATE.md ? Reusable skeleton for a **tracked project doc**: status header · decisions-locked · progress tracker (?/?/?) · definition-of-done · archived on completion. Standardizes the emergent pattern in `RALLY_QUALITY_RESEARCH.md` §7 and `GOLDEN_REGRESSION_FIXTURES.md`. Use for any substantial design/project work.
41:- CODE_CLEANUP_AUDIT_2026-06-24.md ? **? Tracked project (DRAFT, owner-gated):** the next-wave cleanup project ? 33 audit findings across bugs, tech debt, operational risks, and quick wins. Phase 1 = pure refactorings (typed-config migration, app-factory, segmenter validator fix); Phase 2+ = logic fixes. §7 progress tracker is the source of truth; sibling of the completed hardening sweep above.
42:- DECISION_SNAPSHOT_REGRESSION.md ? **? Tracked project (DRAFT, owner-gated ? decisions in §2 need locking):** a deterministic **quality-neutral proof** ? freeze + compare the pipeline's **decision output** (rally segments + stitch/concat plan) off a cached trajectory, GPU-free, in CI. The behavioral proof that lets a *genuine* (non-AST-identical) refactor be confidently called quality-neutral. **Never hashes the encoded reel** (non-deterministic). Extends the existing characterization snapshot (`test_golden_fixtures.py`) from segmentation to the stitch plan; peer of the M4 quality-floor + GPU `nightly_reelcount.py`. §7 tracker is the source of truth.

```
=== Total docs/*.md count and archives dir layout ===
docs/archives/checkpoints/
docs/archives/decisions/
docs/archives/field-pmf/
docs/archives/past_projects/
docs/archives/quality-iterations/
docs/archives/research/
docs/archives/roora-vendor/
---
DECOUPLED_COMPUTE
README.md
low-regret-moves
```

23. assistant (2026-06-29T17:03:00.242Z)

I now have everything needed. Let me confirm the README does NOT index FIXES_PLAN or CODE_REVIEW (confirmed ? neither appears in README.md index), which means FIXES_PLAN's own header claim ("indexed in docs/README.md") is itself slightly stale, and there's no README inbound to update. Let me do one final check: confirm there are no other inbound refs to FIXES_PLAN/CODE_REVIEW elsewhere (memory files, other .md), and verify the CODE_REVIEW doc's own status/terminal nature.

24. user (2026-06-29T17:03:07.366Z)

```
=== ALL refs to FIXES_PLAN or CODE_REVIEW anywhere under repo (md + any) ===
docs/RALLY_DETECTION_FIXES_PLAN.md:129:- `docs/RALLY_DETECTION_CODE_REVIEW.md`
docs/RALLY_DETECTION_CODE_REVIEW.md:9:> `sports-data-collector` repo. See
`docs/RALLY_DETECTION_FIXES_PLAN.md` §7/§8 for status + the
docs/NEXT_STEPS.md:25:9b. **[owner · GPU golden] Promote the R4 density fix (finding B / P2 in
`RALLY_DETECTION_FIXES_PLAN.md`).** PR #393 landed it behind `inactivity_temporal_density`
(**default-OFF**, zero-regression no-op in prod). To ship: golden-set A/B with the flag ON vs
OFF at the **SERVED** windowing (the eval `WindowingPreset` carries it); flip the config default
to True in a one-line PR only if tolF1 doesn't regress. Fixes a sparse-tracking quirk where one
fast sample after a detection gap held the rally tail open (R4 is default-ON in prod).

=== CODE_REVIEW.md: any status header / lifecycle line? ===
9:> `sports-data-collector` repo. See `docs/RALLY_DETECTION_FIXES_PLAN.md` §7/§8 for status +
the

=== Does CODE_REVIEW reference QUALITY_REPORT (lineage)? FIXES_PLAN §10 references CODE_REVIEW
===
9:> `sports-data-collector` repo. See `docs/RALLY_DETECTION_FIXES_PLAN.md` §7/§8 for status +
the
```

25. assistant (2026-06-29T17:03:17.526Z)

I now have a complete and verified picture. Let me synthesize the assessment.

Key findings:
- **RALLY_DETECTION_FIXES_PLAN.md** is genuinely terminal (Status: DONE, all 3 findings A/B/C resolved, P2b merged #396). BUT its own header (line 8, line 30) explicitly states "archival deferred until the wider R-series quality cluster is terminal, per docs/DOC_STATUS.md." It is part of the gated R-cluster (it's the FIXES sub-track of rally detection quality, cross-linked to QUALITY_REPORT and CODE_REVIEW).
- **The cluster gate (DOC_STATUS §4 finding #3) is NOT closed by P2b.** The gate has two prongs, both still open: (1) R2 tolF1 promoted-or-rejected as the gate via the ablation experiment ? still pending (NEXT_STEPS.md line 21 item 8 lists it as owner-side GPU TODO; R2 doc §6/§7 still says tIoU is the gate, R2 augments until ablation); (2) §7.1 single quality number attributed ? RALLY_QUALITY_RESEARCH §7.2 shows Tier-1/2 items (W1.5, W2, M1, M2) still ?/? and §7.1 gap-closure % is TODO. P2b promoted the `inactivity_temporal_density` operational flag (a do-no-harm correctness fix with NO measured lift, CI spans 0) ? this is a different flag entirely from the R2 tolF1 metric-gate ablation.

So NOTHING in the R-cluster is archive-ready yet, including FIXES_PLAN (it's gated to the cluster by its own header + DOC_STATUS policy).

26. user (2026-06-29T17:03:55.866Z)

Structured output provided successfully